PLAN: AI-Skill Scout – OpenClaw Agent

> For the 1-page overview, see **HLD.md**.

Status: Draft v4 **Date:** 2026-04-15 **Scope:** An OpenClaw agent that automatically searches GitHub for high-quality OpenClaw skills related to developing AI models, then installs them safely.

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1. Goal

Build an **OpenClaw agent** (`openclaw agents add ai-skill-scout`) that, on demand:

1. Searches GitHub for existing OpenClaw skills (SKILL.md files) relevant to AI model development
2. Evaluates quality using free signals (repo trust, skill completeness, security scan)
3. Installs the best match safely (quarantine -> scan -> user approval -> install)
4. Reports what was installed and what gaps remain

The search is **open-ended** – not a fixed list. The linked LLM (model-agnostic) expands queries with synonyms. The agent works with any model configured in OpenClaw.

Current Landscape

78 ML skills exist on GitHub but are scattered. The tools practitioners use most (Unsloth, Axolotl, TRL, lm-eval-harness, W&B, DVC) have zero existing skills. No agent currently finds and installs the right skill on demand.

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2. Architecture – 7-Step Pipeline (modeled on Hermes Agent)

```
User: "I need a fine-tuning skill"
  |
  v
[1. CHECK] -- Already installed?
  |  openclaw skills list | grep query
  |  Check installed-skills.json
  |
  v
[2. SEARCH] -- GitHub search with LLM query expansion
  |  Check search-cache.json (1hr TTL)
  |  LLM expands: "fine-tuning" -> lora, qlora, sft, peft, unsloth, axolotl...
```

```
  |  gh search code --filename SKILL.md "<expanded queries>"
  |  gh search repos "SKILL.md <query>"
  |  On 429/timeout -> fall back to cached results
  |
  v
[3. EVALUATE] -- Rank by trust + completeness
  |  Trust: well-known orgs (HIGH) > 1K+ star repos (MEDIUM) > community (LOW)
  |  Completeness: frontmatter, triggers, install steps, CLI commands?
  |  Dedup against installed skills
  |  Classify: ADOPT (exact match) / EXTEND (partial) / GAP (nothing found)
  |
  v
[4. PRESENT] -- Top 3-5 candidates with trust signals
  |  User approves which to install (never auto-install)
  |
  v
[5. QUARANTINE + SCAN] -- Download to temp, security scan
  |  Download to /tmp/skill-quarantine/<name>/
  |  Regex scan (50+ patterns): exfiltration, injection, destructive,
  |    obfuscation, credential access
  |  Trust-level policy: HIGH=allow caution, MEDIUM=ask, LOW=block caution
  |
  v
[6. INSTALL] -- Place in workspace + verify + log
  |  Move to ~/.openclaw/workspace/skills/<name>/SKILL.md
  |  openclaw skills check
  |  Write to installed-skills.json + skill-audit.log
  |
  v
[7. REPORT] -- Confirm install + log gaps
     "Installed peft (HIGH trust). No skill found for Unsloth (GAP)."
```

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3. Detailed Design

3.1 Search

Source: GitHub only. Two search methods:

```
gh search code --filename SKILL.md "fine-tuning OR lora OR training" --limit 20
gh search repos "SKILL.md fine-tuning" --limit 20
```

Query expansion: LLM expands user query to cover synonyms ("fine-tuning" -> lora, qlora, sft, peft, unsloth, axolotl, trl, instruction-tuning). Limited to 3-5 queries per request.

Caching: Results cached in `search-cache.json` with 1hr TTL. On GitHub

429/timeout, return cached results with staleness notice.

3.2 Evaluation

Signal	Weight	Source
Repo trust (known org / stars / community)	Highest	GitHub API (free)
Skill completeness (frontmatter, triggers, CLI commands)	High	LLM analysis
Repo health (stars, contributors, last push)	Medium	GitHub API (free)
Deduplication (overlap with installed skills)	Pass/fail	Local check

3.3 Security (Quarantine + Scan)

After user approves a candidate:

1. **Quarantine:** Download to `/tmp/skill-quarantine/<name>/` – never directly to workspace
2. **Scan:** Regex-based static analysis:

Category	Example patterns	Action
Exfiltration	`curl.*|`, `wget.*POST`, base64+send	BLOCK
Injection	`eval(`, `exec(`, `os.system(`	BLOCK
Destructive	`rm -rf`, `DROP TABLE`, `mkfs`	BLOCK
Obfuscation	base64-encoded commands, hex URLs	BLOCK
Credential access	`cat ~/.ssh`, `$AWS_SECRET` exfil	BLOCK
Caution	External API calls, non-PyPI pip install	Trust-dependent

3. **Trust-level policy:**

Trust Level	Safe	Caution	Dangerous
HIGH (known orgs)	Allow	Allow with notice	Block
MEDIUM (1K+ stars)	Allow	Ask user	Block
LOW (community)	Allow	Block	Block

3.4 Install + Tracking

```
mkdir -p ~/.openclaw/workspace/skills/<name>/
cp /tmp/skill-quarantine/<name>/SKILL.md ~/.openclaw/workspace/skills/<name>/
openclaw skills check
```

Lockfile (`installed-skills.json`):

```json
{
  "peft": {
    "source": "github:huggingface/skills/skills/peft",
    "installed_at": "2026-04-15T10:30:00Z",
    "trust_level": "HIGH",
    "content_hash": "sha256:abc123...",
    "scan_result": "clean"
  }
}
```

Audit log (`skill-audit.log`):

```
2026-04-15T10:30:00Z INSTALL peft source=github:huggingface/skills trust=HIGH scan=clean
2026-04-15T10:31:00Z SEARCH "fine-tuning" results=5 installed=1 gaps=["unsloth"]
```

Gap tracking (`gaps.json`): Tools the user asked for but no skill exists.

3.5 Scheduled Refresh

```bash
openclaw cron add --schedule "0 9 * * 1" \
  --prompt "Search GitHub for new ML/AI OpenClaw skills. Compare against gaps.json. Report r
```

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4. What the Agent Searches For

Any OpenClaw skill relevant to developing, training, and evaluating AI models. Open-ended, not a fixed list. Core areas:

Area	Example Tools	Existing Skills?
Fine-tuning (LoRA, QLoRA, SFT)	Unsloth, Axolotl, TRL, LLaMA-Factory, torchtune	`peft`, `fine-tuning` exist; Unsloth/Axolotl/TRL = GAP
Alignment (DPO, GRPO, RLHF)	TRL, OpenRLHF	Only via HF trainer
Evaluation	lm-eval-harness, DeepEval, Promptfoo	`promptfoo` exists; lm-eval = GAP
Serving & inference	Ollama, vLLM, SGLang, llama.cpp	`ollama-local`, `vllm` exist
Quantization (GGUF, AWQ)	llama.cpp, AutoAWQ	GAP

Area	Example Tools	Existing Skills?
Data preparation	distilabel, Argilla, Label Studio	GAP
Experiment tracking	W&B, MLflow	`mlflow` basic exists; W&B = GAP
Data versioning	DVC	GAP
Hyperparameter tuning	Optuna	GAP
MLOps	dstack, BentoML	`dstack` exists

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5. Implementation Checklist

Phase 1: Discover & Install

- ☐ Create agent: `openclaw agents add ai-skill-scout --workspace ~/ai-skill-scout`
- ☐ Configure AGENTS.md with agent identity and behavior
- ☐ Register tools: `find_ai_skill(query)`, `install_skill(url)`, `list_installed_ml_skills()`
- ☐ Implement GitHub search with LLM query expansion
- ☐ Implement evaluation (trust + completeness ranking)
- ☐ Implement quarantine + security scan pipeline
- ☐ Implement install with lockfile + audit log
- ☐ Implement search cache (1hr TTL) + graceful degradation
- ☐ Implement gap tracking
- ☐ Test: agent responds to "I need a fine-tuning skill" end-to-end

Phase 1b: Continuous Discovery

- ☐ Set up weekly cron via `openclaw cron add`
- ☐ Weekly search compares against gaps.json
- ☐ Alert user to new high-quality skills found

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6. Stack

Component	Technology
Runtime	OpenClaw 2026.4.12 (`openclaw agents add`)
LLM	Any linked model (model-agnostic)
Search	`gh search code` + `gh search repos`
Install	`curl` download + workspace placement

Component	Technology
Scheduling	`openclaw cron add`
Cache	JSON files in agent workspace

No external dependencies beyond `openclaw`, `gh`, and `curl` (all already installed).

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7. Open Questions

#	Question	Resolution Path
1	Where exactly should workspace skills go?	Verify `~/.openclaw/workspace/skills/` is correct
2	How to handle skills with supporting files (not just SKILL.md)?	Download full skill directory from GitHub
3	Should gaps.json be shared across sessions?	Store in agent workspace for persistence

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8. References

Research (informing design patterns)

Pattern	Source	What We Borrow
Skill search + install pipeline	Hermes Agent (86K stars)	Quarantine-scan-confirm-install, trust levels, audit log
Progressive disclosure	Agent Skills Survey (arXiv:2602.12430)	L1 metadata -> L2 detail -> L3 full content
Security scanning	Hermes skills_guard.py	50+ regex threat patterns, trust-level policy matrix
Query expansion	Gorilla LLM (12K stars)	Retriever-aware mapping of user needs to tools
Decision framework	claude-skill-search-first	ADOPT / EXTEND / GAP classification
Skill format	OpenClaw SKILL.md spec	YAML frontmatter + metadata.openclaw + markdown body

Pattern	Source	What We Borrow
Community skill safety	“Six Million Fake Stars” (ICSE 2026)	26.1% vulnerability rate; never auto-install

Key Repos

Project	Stars	Relevance
Hermes Agent	86K	Reference implementation for skill search/install pipeline
OpenClaw	357K	Platform, SKILL.md format, Plugin SDK
awesome-openclaw-skills	46K	Curated skill catalog, quality filtering
AI-Build-AI	243	Target project for AI model development

OpenClaw Docs

- Skills: https://docs.openclaw.ai/tools/skills
- Creating Skills: https://docs.openclaw.ai/tools/creating-skills
- Plugin SDK: https://docs.openclaw.ai/plugins/building-plugins